

Homework 5

Problem 1. Let a_n be the number of ordered triples $\langle i, j, k \rangle$ of integer numbers such that $i \geq 0, j \geq 1, k \geq 1$, and $i + 3j + 3k = n$. Find the generating function of the sequence $(a_0, a_1, a_2, \dots)$ and calculate a formula for a_n .

Solution.

$$\begin{aligned} & (1 + x + x^2 + x^3 + \dots)(x^3 + x^6 + x^9 + \dots)(x^3 + x^6 + x^9 + \dots) \\ &= \frac{1}{1-x} \frac{x^3}{1-x^3} \frac{x^3}{1-x^3} \\ &= \frac{x^6(1+x+x^2)}{(1-x^3)^3} = x^6(1+x+x^2)(1-x^3)^{-3}. \end{aligned}$$

Then use the generalized binomial theorem. □

Problem 2. If $a(x)$ is the generating function of a sequence $(a_0, a_1, a_2, \dots)$, please find the generating function of the sequence of partial sums $(a_0, a_0 + a_1, a_0 + a_1 + a_2, \dots)$.

Solution.

$$\begin{aligned} & a_0 + (a_0 + a_1)x + (a_0 + a_1 + a_2)x^2 + (a_0 + a_1 + a_2 + a_3)x^3 \dots \\ &= a_0(1 + x + x^2 + \dots) \\ & \quad + a_1x(1 + x + x^2 + \dots) \\ & \quad + a_2x^2(1 + x + x^2 + \dots) \\ &= \frac{a_0}{1-x} + \frac{a_1x}{1-x} + \frac{a_2x^2}{1-x} + \dots \\ &= \frac{1}{1-x}a(x) \end{aligned}$$

□

Problem 3. Express the n^{th} term of the sequences given by the following recurrence relations

1. $a_0 = 2, a_1 = 3, a_{n+2} = 3a_n - 2a_{n+1} \ (n = 0, 1, 2, \dots)$.
2. $a_0 = 1, a_{n+1} = 2a_n + 3 \ (n = 0, 1, 2, \dots)$.

Solution.

1. Characteristic function is $x^2 + 2x - 3 = (x + 3)(x - 1) = 0$.

$$\text{Let } f_n = a(-3)^n + b \cdot 1^n. \text{ Then } \begin{cases} 2 &= a + b \\ 3 &= -3a + b \end{cases} \Rightarrow a = -1/4, b = 9/4.$$

$\therefore$ the n -th term is f_n .

2. Characteristic function for the homogeneous part is $x = 2$. Take $a_n = p2^n + \lambda$

$$a_0 = 1, a_1 = 5. \text{ Now } \begin{cases} 1 &= p + \lambda \\ 5 &= 2p + \lambda \end{cases} \Rightarrow p = 4, \lambda = -3.$$

□

Problem 4. Solve the recurrence relation $a_{n+2} = \sqrt{a_{n+1}a_n}$ with initial conditions $a_0 = 2, a_1 = 8$ and find $\lim_{n \rightarrow \infty} a_n$.

Solution. Consider the sequence $b_n = \log_2 a_n$. Then

$$2 \log_2 a_{n+2} = \log_2 a_{n+1} + \log_2 a_n$$

i.e. $2b_{n+2} = b_{n+1} + b_n$. $b_0 = 1, b_1 = 3$. One can find $b_n = (-\frac{4}{3})(-\frac{1}{2})^n + \frac{7}{3}$.
 $\therefore a_n = 2^{(-\frac{4}{3})(-\frac{1}{2})^n + \frac{7}{3}}$. $\lim_{n \rightarrow \infty} a_n = 2^{\frac{7}{3}}$. □